

polytetrafluoroethylene (PTFE), CEM-1 through CEM-5, G-10 and G-11, RF-35, polyimide, alumina, and flexible substrates like Pyralux and Kapton.

Shavinder Singla, technical officer, Centre for Development of Advanced Computing, says, "IMS takes care of heat dissipation created by modern PCB designs, which are often linked to interconnection difficulties. Care must be taken while selecting IMSes for an application."

"In general, IMSes minimise thermal impedance and conduct heat more effectively. These substrates are mechanically more robust than thick-film ceramics and direct-bond copper constructions that are often used in many applications."

Laminate. This provides properties like coefficient of thermal expansion, tensile and shear strength, and Tg. Common dielectrics used for laminates are CEM-1 and CEM-3, FR-1, FR-4, polytetrafluoroethylene (teflon), FR-2 through FR-6, CEM-1 through CEM-5, and G-10.

Copper foil is the next layer that is laminated to the board. For double-sided PCBs, copper is applied to both sides of the substrate. Copper's thickness varies as per the application. For example, thickness is more for high-power applications, as compared to low-power ones.

Solder mask. This is the layer on top of the copper foil. It works as an insulation for copper traces from accidental contact with other conductive metals. It helps soldering at correct places.

It is a protective layer that guards against external contaminants and provides required isolation between surface elements like pads, copper traces and drill holes.

Silkscreen. Overlaid on top of the solder mask, a silkscreen is used to add letters, numbers and symbols to the PCB for easy assembly, and indicators to better understand the board.

Design technologies for PCBs

Each board serves a unique purpose for a particular product. Hence, it must be designed to perform that function in the allotted space. Designers use computer-aided design (CAD)

systems with special software to lay out the circuit pattern on the board. Software available for PCB designs include Allegro PCB Designer, Altium Designer and PCB Droid. Use of software ensures that the design does not contain errors such as traces that are skinny or incorrectly touch, or drill holes that are tiny.

Ajay Singha, principal engineer, Lambda Consulting, opines, "Two most common design technologies for PCB designs are through-hole technology and surface mount technology (SMT). In the former, each component is pressed into tiny holes of the substrate and soldered to connect on the reverse side of the board."



Fig. 2: Multi-layer PCB (Credit: www.elprocus.com)

"In SMT, L-shaped or J-shaped terminals of each component get in touch with the PCB directly."

"Components on a PCB are electrically connected to the circuit(s) by these methods. Although SMT eliminates the time-consuming drilling process and space-consuming connection pads inherent with through-hole technology, greater care is needed in the placement of components."

An IC performs functions that are similar to those of a PCB, except that the IC contains many more circuits and components that are electrochemically grown in place on the surface of a tiny silicon chip.

A hybrid circuit contains some components that are grown onto the surface of the substrate rather being placed on the surface and soldered. It looks like a PCB.

Types of PCBs

Circuit boards may be classified in several ways, as follows:

- Component location: single-

sided, double-sided and embedded

- Stack-up: single layer and multi-layer
- Design: module-based, custom and special
- Bendability: rigid, flex and rigid-flex
- Strength: electrically strong and mechanically strong
- Electrical functionality: high frequency, high power, high density and microwave

The board types can be used to select the circuit board material best suited for the design.

Single-sided PCBs include only one layer of a substrate, coated with a thin layer of copper. A protecting solder mask is laid down onto the copper layer. The silkscreen coat may be applied to the top to mark the elements of the board.

A double-sided PCB's substrate includes metal conductive layers and elements attached to both sides (top and bottom).

Multi-layer PCBs enlarge the density and complexity of PCB designs by adding extra layers beyond the layers seen in the configuration of double-sided PCBs. These allow for extremely thick and highly-compound designs. Extra layers used are power planes that provide the supply to the circuit with power and decrease the levels of electromagnetic interference (EMI).

Rigid PCBs use a solid, rigid substrate material like fibre-glass that keeps the board from twisting. A motherboard within a computer is the best example of an inflexible PCB.

A flex PCB's substrate is flexible plastic. It enables turning and shifting during use without harming the circuit(s) on the PCB. It can restore heavy wiring in superior gear where weight and space matter, such as satellites.

Rigid-flex boards consist of a rigid circuit board attached to a flex circuit board. These boards can fulfill compound design requests on demand.

Safety considerations for toxic materials. Fumes from the solder may contain lead, which is toxic. Solder is used to make electrical connections on a PCB, so soldering operations must be carried out in a